Random holdout

Temporal holdout

All substitutions

De novo only

A

antiGen (multi−tree) 0.86
antiGen (single−tree) 0.83
Where x which 0.79
Frequency (memorisation) 0.79
Substitution matrix 0.70
Where only 0.50
Codon accessibility 0.32
Uniform 0.10

0.00 0.25 0.50 0.75
Recall at top 10%

B

antiGen (multi−tree) 0.81
antiGen (single−tree) 0.77
Where x which 0.73
Frequency (memorisation) 0.72
Substitution matrix 0.65
Where only 0.44
Codon accessibility 0.30
Uniform 0.10

0.00 0.25 0.50 0.75
Recall at top 10%

C

antiGen (multi−tree) 0.45
antiGen (single−tree) 0.36
Where x which 0.45
Frequency (memorisation) 0.05
Substitution matrix 0.46
Where only 0.17
Codon accessibility 0.33
Uniform 0.10

0.00 0.25 0.50 0.75
Recall at top 10%

D

antiGen (multi−tree) 0.51
antiGen (single−tree) 0.39
Where x which 0.50
Frequency (memorisation) 0.05
Substitution matrix 0.47
Where only 0.18
Codon accessibility 0.34
Uniform 0.10

0.00 0.25 0.50 0.75
Recall at top 10%

All substitutions · De novo only · 95% interval excludes zero · interval includes zero

Versus single−tree

E

SARS−CoV−2 S
H5Nx HA
Measles
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled

0.0 0.2 0.4 0.6 0.8
Multi−tree minus comparator, recall at top 10%

F

SARS−CoV−2 S
H5Nx HA
Measles
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled

−0.4 0.0 0.4
Multi−tree minus comparator, recall at top 10%

Versus where x which

G

SARS−CoV−2 S
H5Nx HA
Measles
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled

−0.2 0.0 0.2 0.4
Multi−tree minus comparator, recall at top 10%

H

SARS−CoV−2 S
H5Nx HA
Measles
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled

−0.4 0.0 0.4
Multi−tree minus comparator, recall at top 10%